

VO5 Virtual Mobile Applications

Description & Eligibility

Contestants will develop a mobile phone and/or tablet application based upon the given scenario. Permitted platforms are limited to Google Android™ or Apple iOS™. This national event will be submitted and judged virtually. Any middle level, secondary or postsecondary contestant may enter this event. There are no restrictions on the number of entries per chapter or per state. Contestants who do not submit an entry following the topic listed below or do not follow the Copyright and/or Fair Use Guidelines and the Artificial Intelligence Usage and Citation policy will be disqualified.

Topic

SafelyGo – Student Campus Safety App Develop a mobile application titled SafelyGo. Student Campus Safety App. You are challenged to create a mobile app that promotes student safety on and around campus. The app should allow users to report safety concerns, access emergency contacts, share real-time location with trusted contacts, and view a live campus safety map.

Required Features: external backend database, user registration and account recovery, ability to submit and view safety reports, emergency contact directory, location sharing, and an admin panel for campus safety staff to manage reports. Must use at least one API (e.g., maps or SMS/notification service).

Competition Notes

- AI Policy Level 2 *Permitted with Citation*
- Contest Registration
 - Advisors register contestants for the event using the Membership Registration System, accessible at <https://register.bpa.org>
 - Click on the “Conferences” tab and then select the “2027 National Virtual Events”.
 - There will be a \$10 fee for each entry (*invoiced during event registration*)
 - Event registration can be purchased with your original invoice or with a new invoice after member registration
 - The registration deadline is no later than 11:59 p.m. Eastern Time, on December 1, 2026
 - Contestants participating in the competition must be registered for the event prior to the submission deadline to be eligible

Technical Pre-Submit Specifications

What to Submit at https://upload.bpa.org	Saved File Name(s)	National Deadline
Submit the following in one combined PDF: 1. A clickable URL to the mobile application, written description of the project and details of a design concept, source code, and application screenshots 2. Works Cited 3. BPA Release Form(s) 4. AI Usage and Documentation Form	VO5-ContestantID.pdf	January 15, 2027, 11:59 p.m. Eastern Time

Contest Competencies

1. Implement system analysis and design concepts to plan, structure, and guide the architecture of a mobile application
2. Develop a comprehensive project plan and timeline to manage all phases of the mobile application development lifecycle
3. Demonstrate knowledge of program implementation, testing, and deployment processes, including strategies for identifying and resolving technical issues

4. Apply Google Android® or Apple iOS® platform-specific programming languages, tools, and techniques to develop a functional mobile application
5. Demonstrate proficiency in core programming language concepts – including object-oriented, structured, and procedural techniques – applying variables, assignment statements, mathematical operators, decision-making controls, and iterative structures to implement application logic
6. Design a graphical user interface (GUI) for a mobile device by applying design and color principles, incorporating user input controls, icons, images, and other graphic elements to create an intuitive and visually consistent user experience
7. Employ database skills and applied programming techniques – including data management, sequential file access, I/O operations, and internal and external function and/or procedure calls – to handle and persist in application data
8. Integrate external Application Programming Interfaces (APIs) to extend application functionality and connect to third-party data sources and services
9. Apply internal programming documentation standards to ensure code clarity, maintainability, and long-term readability

Contest Specifications for this Event

- The application may be cross platform compatible between Google's Android® or Apple iOS®.
- The application does not need to be available for download from a digital-distribution multimedia-content service such as Google Play®, or Apple App Store®.
- Submit the URL to the mobile application including a written description of the project and details of a design concept, source code (as a zipped root folder), application screenshots, along with Works Cited, and BPA Release Form(s) in a combined PDF file to <https://upload.bpa.org> no later than 11:59 p.m. Eastern Time, on January 15, 2027.
- Advisors of qualifying contestants will be contacted to reserve a time for a videoconference presentation before a panel of judges. The dates and times for presentation reservations will be announced via email notification. Contestants will be assigned to sections prior to the preliminary presentation.
- Length of event: no more than three (3) minutes for setup, no more than ten (10) minutes for the presentation, and no more than five (5) minutes for judges' questions.
- Contestants with the highest cumulative technical and presentation scores will be invited to attend the National Leadership Conference to be recognized for their outstanding efforts and to participate in the National Showcase.

Common Specifications for all BPA Judged Events

- Contestant ID is required for all submissions. Materials from non-registered contestants and/or those missing Contestant ID cannot be accepted.
- The contestant is responsible for securing a BPA Release Form(s) from any person, including oneself, whose image or work is used in the project.
- The contestant is responsible for citing all sources, including oneself, on the Works Cited page for any work used on the project. It is the policy of Business Professionals of America to comply with copyright law. The *Style & Reference Manual* contains guidelines for Copyright and/or Fair Use. Contestants will be disqualified for violations of the guidelines.
- Materials must follow the organization's BPA Graphic Standards and make proper use of the BPA logo and/or organization's name. Refer to the BPA Graphic Standards in the *Style & Reference Manual*.
- Judges' comments will be returned digitally through the online judging system at the national level. Materials submitted for technical judging will *not* be returned and will *not* be available at NLC.

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Technical Scoring Rubric

Required Elements (If any question results in a NO, please assign a score of 0)				
Contestant followed the topic.				☐ Y ☐ N
Contestant followed Copyright and/or Fair Use Guidelines and the Artificial Intelligence Usage & Citation Policy.				☐ Y ☐ N
If answered no to either question, please stop scoring and provide a brief reason for the <i>disqualification</i> :				
Submission and Formatting (All points or none are awarded by the technical judge)				
Contestant submitted one (1) combined PDF file containing the mobile application, written description of the project and details of a design concept, source code, application screenshots and Works Cited formatted according to the <i>Style & Reference Manual</i> , BPA Release Form(s) and the AI Usage and Documentation Form.				/10
User Interface	Below Expectations 1-5 points	Meets Expectations 6-10 points	Above Expectations 11-20 points	Points Awarded
Writing Accuracy	Frequent errors in grammar, spelling, punctuation, or usage that interfere with clarity and professionalism.	Minor errors may be present but do not hinder understanding; generally clear and correct language use.	Consistently accurate and polished use of grammar, spelling, punctuation, and usage; enhances clarity and professionalism.	/20
Originality & Innovation	Work lacks originality; ideas are conventional or derivative with minimal innovation.	Work shows some creative thought and uniqueness; moderately innovative and relevant.	Work is highly original, distinctive, and demonstrates exceptional innovation and creativity.	/20
Mobile Optimization	Design is poorly adapted for mobile; elements may be misaligned, hard to read, or not responsive.	Design functions well on mobile; layout, readability, and responsiveness are adequately addressed.	Design is fully optimized for mobile; intuitive layout, excellent readability, and seamless responsiveness.	/20
Visual Design Elements	Design elements are inconsistent or poorly chosen; color, typography, or graphics detract from clarity or appeal.	Design uses color, typography, and graphics appropriately; visually cohesive and supports communication.	Design elements are expertly integrated; color, typography, and graphics enhance aesthetics and elevate the overall impact.	/20
Platform Compliance	Design disregards platform guidelines; may result in usability issues or inconsistent user experience.	Design generally follows platform standards; interface is functional and aligns with expected behaviors.	Design fully complies with platform guidelines; interface is polished, intuitive, and enhances user experience through thoughtful adherence.	/20
Clarity & Simplicity	Interface is confusing or cluttered; layout and elements hinder user understanding.	Interface is generally clear and organized; users can navigate and understand with minimal effort.	Interface is exceptionally intuitive and streamlined; clarity and simplicity enhance usability and user satisfaction.	/20
Code & Data	Below Expectations 1-5 points	Meets Expectations 6-10 points	Above Expectations 11-20 points	Points Awarded
Platform Usage	Application does not run on a permitted platform or lacks proper deployment.	Application is functional on at least one permitted platform (Android or iOS) with basic compatibility.	Application is well-integrated and optimized for one or both permitted platforms, demonstrating thoughtful use of platform-specific features.	/20

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Technical Scoring Rubric (Cont'd)

Code Quality	Code is disorganized and hard to read; lacks consistent formatting, meaningful naming, and logical structure.	Code is generally clear and readable; follows standard conventions and is logically organized.	Code is clean, well-structured, and follows best practices for readability and maintainability.	/20
Object-Oriented Design	Code shows limited or incorrect use of OOP concepts; design patterns are absent or misapplied.	Code applies basic OOP principles appropriately; uses common design patterns where relevant.	Code shows strong OOP skills with thoughtful use of design patterns for structure and scalability.	/20
Database Integration	No database integration or improper use; data is hardcoded or handled inefficiently.	Code connects to and retrieves data from a database; basic queries and data handling are implemented correctly.	Code integrates the database effectively with secure handling, efficient queries, and well-designed schemas.	/20
Code Documentation	Documentation is missing, incomplete, or unclear; code is difficult to understand without external explanation.	Code includes adequate comments and documentation; key functions and logic are explained.	Code is clearly documented; comments support readability, maintainability, and collaboration.	/20
User Functionality	Below Expectations 1-5 points	Meets Expectations 6-10 points	Above Expectations 11-20 points	Points Awarded
Device Deployment	Application does not run or deploy on a permitted mobile device; platform compatibility is missing or incomplete.	Application successfully runs or deploys on at least one permitted mobile device; basic functionality is intact.	The application is fully deployed and optimized for mobile, with smooth performance and seamless integration.	/20
Database Functionality	Application fails to connect to or retrieve data from the database; functionality is broken or unreliable.	Application successfully loads and retrieves data from the database; basic queries function as intended.	Application reliably and efficiently interacts with the database; data retrieval is accurate, fast, and supports user needs seamlessly.	/20
Project Plan	Below Expectations 1-5 points	Meets Expectations 6-10 points	Above Expectations 11-20 points	Points Awarded
Writing Accuracy	Frequent errors in grammar, spelling, punctuation, or usage that interfere with clarity and professionalism.	Minor errors may be present but do not hinder understanding; generally clear and correct language use.	Consistently accurate and polished use of grammar, spelling, punctuation, and usage; enhances clarity and professionalism.	/20
Project Planning	Plan is vague or incomplete; lacks clear steps, timelines, or role assignments to meet project goals.	Plan adequately details tasks, timelines, and responsibilities; shows a reasonable path to meeting requirements.	Plan is thorough, well-structured, and proactive; includes detailed milestones, contingency strategies, and clear accountability to ensure timely completion.	/20
TOTAL TECHNICAL POINTS				/310

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Presentation Scoring Rubric

Required Elements (If any question results in a NO, please assign a score of 0)				
Contestant followed the topic.				☐ Y ☐ N
Contestant followed Copyright and/or Fair Use Guidelines and the Artificial Intelligence Usage & Citation Policy.				☐ Y ☐ N
If answered no to either question, please stop scoring and provide a brief reason for the <i>disqualification</i> :				
Evaluation Criteria	Below Expectations 1-5 points	Meets Expectations 6-10 points	Above Expectations 11-20 points	Points Awarded
Development/Design Process	Explanation is unclear or incomplete; lacks detail about the development or design process.	Contestant adequately describes the development and design process; includes key decisions and steps taken.	Contestant provides a clear, thorough, and insightful explanation of the development and design process; demonstrates deep understanding and thoughtful reflection.	/20
Application Demonstration	Demonstration is unclear or incomplete; user interface is not effectively used to show functionality.	Contestant adequately demonstrates key features through the user interface; audience can follow how the application works.	Demonstration is clear, engaging, and thorough; user interface is used effectively to showcase functionality and user experience.	/20
Logical Design & Patterns	Explanation is unclear or lacks detail; logical design and design patterns are not addressed or misunderstood.	Contestant adequately explains the application's structure and identifies relevant design patterns used.	Contestant provides a clear, insightful explanation of the logical design and thoughtfully discusses the use and benefits of design patterns.	/20
Code Functionality	Explanation is vague or incorrect; presenter struggles to connect code structure to application functionality.	Explains how code structures support features; shows basic understanding.	Confidently explains code structures and their role in functionality and design.	/20
Interview Responses	Responses are unclear, incomplete, or off topic; presenter struggles to address questions effectively.	Contestant responds clearly and directly to most questions; demonstrates understanding of the project and communicates effectively.	Contestant responds confidently and thoughtfully, showing deep understanding and engaging with questions insightfully.	/20
Presentation Style	Presentation is disorganized or difficult to follow; delivery lacks clarity, confidence, or professionalism.	Presentation is clear and adequately structured; delivery is generally confident and appropriate.	Presentation is clear, engaging, and professionally delivered to support audience understanding.	/20
Timing	Setup lasted no more than three (3) minutes, and the presentation lasted no more than ten (10) minutes.			/10
TOTAL PRESENTATION POINTS				/130

Props and/or additional items shall not be used as a basis for scoring.

PRESENTATION WILL BE STOPPED AT TEN (10) MINUTES